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Subject Name: Database Management System **Subject Code: 24CSH-298**

WORKSHEET 5

AIM: To understand and apply conditional logic in SQL using the MOD (%) operator to classify employee salaries as odd or even.

S/W Requirement:
- Database Management System: PostgreSQL / Oracle Database Express Edition
- Database Administration Tool: pgAdmin

OBJECTIVES:

To write and execute SQL queries that:

- Use the MOD (%) operator
- Identify whether salary values are odd or even
- Display employee details based on salary classification

PROBLEM STATEMENT:

Develop and execute SQL queries that utilize conditional logic to analyze employee salary data. The queries should determine whether salaries are odd or even using the MOD operator and display appropriate results.

```
CREATE TABLE Employee2 (
    emp_id NUMERIC,
    emp_name VARCHAR(50),
    salary NUMERIC
);
```

1. **PROBLEM STATEMENT – Display Salary Type (Odd/Even)**

Write an SQL query to display employee details along with salary classification as odd or even.

PROGRAM:

```sql
SELECT emp_id, salary,
       CASE
           WHEN MOD(salary, 2) = 0 THEN 'Even'
           ELSE 'Odd'
       END AS salary_type
FROM Employee2;
```

OUTPUT:

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	emp_id numeric	salary numeric	salary_type text
1	1	20000	Even
2	2	0	Even
3	3	25000	Even
4	4	56011	Odd
5	5	24013	Odd

2. **PROBLEM STATEMENT – Display Employees with Even Salaries**

Write an SQL query to display employees whose salaries are even.

PROGRAM:

```sql
SELECT *
FROM Employee2
WHERE MOD(salary, 2) = 0;
```

OUTPUT:

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	emp_id numeric	emp_name character varying (50)	salary numeric
1	1	Rahul	20000
2	2	Amit	0
3	3	Neha	25000

3. PROBLEM STATEMENT – Display Employees with Odd Salaries

Write an SQL query to display employees whose salaries are odd.

PROGRAM:

```
SELECT *
FROM Employee2
WHERE MOD(salary, 2) <> 0;
```

OUTPUT:

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	emp_id numeric	emp_name character varying (50)	salary numeric
1	4	Jay	56011
2	5	Sushmita	24013

4. PROBLEM STATEMENT – Count Odd and Even Salaries

Write an SQL query to count employees based on salary type.

PROGRAM:

```sql
SELECT
    CASE
        WHEN MOD(salary, 2) = 0 THEN 'Even'
        ELSE 'Odd'
    END AS salary_type,
    COUNT(*) AS total_employees
FROM Employee2
GROUP BY
    CASE
        WHEN MOD(salary, 2) = 0 THEN 'Even'
        ELSE 'Odd'
    END;
```

OUTPUT:

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	salary_type text	total_employees bigint
1	Even	3
2	Odd	2

LEARNING OUTCOMES:

1. Learned how to use the MOD (%) operator in SQL
2. Understood classification of data into odd and even values
3. Gained knowledge of CASE statements in SQL
4. Improved data filtering and grouping techniques
5. Enhanced SQL query writing skills

CONCLUSION:

This experiment demonstrated how conditional logic can be implemented in SQL using the MOD operator. It helped in classifying and analyzing employee salary data effectively, improving decision-making and data handling skills.